

Introduction to Scikit-learn

- Scikit-learn
 - Machine learning library built on **Numpy** and **Matplotlib**
- What Scikit-learn can do
 - **Unsupervised** learning
 - Clustering
 - **Supervised** learning
 - Regression, classification
 - Data **preprocessing**
 - Feature extraction, feature selection, dimensionality reduction

Introduction to Scikit-learn

- What Scikit-learn **cannot** do
 - Distributed computation on multiple computers
 - Only multi-core optimization
 - Deep learning
 - Use Keras and Tensorflow instead

Introduction to Scikit-learn

- Scikit learn models work with structured data
 - Data must be in the form of **2D Numpy arrays**
 - Rows represent the **samples**
 - Columns represent the **attributes (or features)**
 - This table is called **features matrix**

shape = (3, 3)



A diagram illustrating a 3x3 features matrix. A blue arrow points from the text 'shape = (3, 3)' to the matrix. The matrix is composed of two parts: a vertical column of labels on the left and a 3x3 grid of numerical values on the right. The labels are 'Sample 1', 'Sample 2', and 'Sample 3'. The numerical values are arranged in three columns labeled 'Price', 'Quantity', and 'Liters' at the top.

	Price	Quantity	Liters
Sample 1	1.0	5	1.5
Sample 2	1.4	10	0.3
Sample 3	5.0	8	1

Introduction to Scikit-learn

- Features can be
 - **Real** values
 - **Integer** values to represent categorical data
- If you have strings in your data, you first have to convert them to integers (**preprocessing**)

Input data

1.0	January	1.5
1.4	February	0.3
5.0	March	1



Features matrix

1.0	0	1.5
1.4	1	0.3
5.0	2	1

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- Also **missing values** must be solved before applying any model
 - With imputation or by removing rows

Input data

1.0	0.5	1.5
1.4	NaN	0.3
5.0	0.5	1

Features matrix

1.0	0.5	1.5
1.4	0.5	0.3
5.0	0.5	1



Input data

1.0	0.5	1.5
1.4	NaN	0.3
5.0	0.5	1

Features matrix

1.0	0.5	1.5
5.0	0.5	1



Introduction to Scikit-learn

- For **unsupervised** learning you only need the features matrix
- For **supervised** learning you also need a **target** array to train the model
 - It is typically one-dimensional, with length `n_samples`

Features matrix
shape = (n_samples, n_features)

1.0	5	1.5
1.4	10	0.3
5.0	8	1

Target array
shape = (n_samples,)

A
A
B

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- The target array can contain
 - Integer values, each corresponding to a class label

Target labels

Dog
Dog
Cat



Target array

0
0
1

- Real values for regression

Target array

0.4
1.8
-6.9

Introduction to Scikit-learn

- Scikit-learn estimator API
 - All models are represented with Python classes
 - Their classes include
 - The values of the **hyperparameters** used to configure the model
 - The values of the **parameters** learned after training
 - By convention these attributes end with an underscore
 - The **methods** to train the model and make inference
 - Scikit-learn models are provided with sensible **defaults** for the hyperparameters

Introduction to Scikit-learn

- Scikit learn models follow a simple, shared **pattern**
 1. **Import** the model that you need to use
 2. **Build** the model, setting its hyperparameters
 3. **Train** model parameters on your data
 - Using the fit method
 4. **Use** the model to make predictions
 - Using the predict/transform methods
- Sometimes fit and predict/transform are implemented within the same class method

Introduction to Scikit-learn

- **fit():** learn model parameters from input data
 - E.g. train a classifier
- **predict():** apply model parameters to make predictions on data
 - E.g. predict class labels
- **fit_predict():** fit model and make predictions
 - E.g. apply clustering to data
- **fit_transform():** fit model and transform data
 - E.g. apply PCA to transform data